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Constructors in Python

In object-oriented programming, **A constructor is a special method used to create and initialize an object of a class.** This method is defined in the class.

- The constructor is executed automatically at the time of object creation.
- The primary use of a constructor is to declare and initialize data member/instance variables of a class. The constructor contains a collection of statements (i.e., instructions) that executes at the time of object creation to initialize the attributes of an object. **Syntax** of a constructor

```
def __init__(self):  
    # body of the constructor
```

Where,

- **def:** The keyword is used to define function.
- **__init__()** Method: It is a reserved method. This method gets called as soon as an object of a class is instantiated.
- **self:** The first argument **self** refers to the current object. It binds the instance to the **__init__()** method. It's usually named **self** to follow the naming convention.

Note: The **__init__()** method arguments are optional. We can define a constructor with any number of arguments.

Types of Constructors

In Python, we have the following three types of constructors.

- Default Constructor
- Non-parameterized constructor
- Parameterized constructor



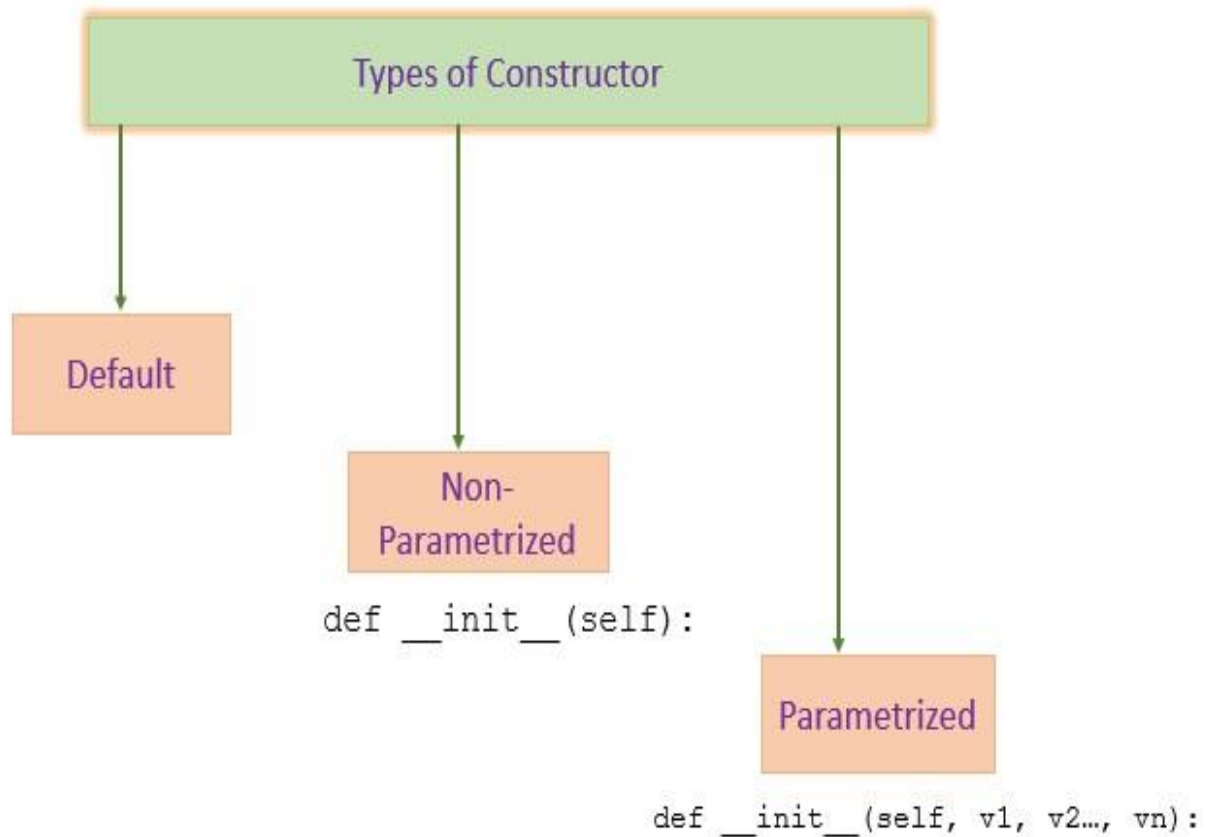
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Default Constructor

Python will provide a default constructor if no constructor is defined. Python adds a default constructor when we do not include the constructor in the class or forget to declare it. It does not perform any task but initializes the objects. It is an empty constructor without a body.

If you do not implement any constructor in your class or forget to declare it, the Python inserts a default constructor into your code on your behalf. This constructor is known as the default constructor.

It does not perform any task but initializes the objects. It is an empty constructor without a body.



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Example:

```
class Employee:
    def display(self):
        print('Inside Display')
emp = Employee()
emp.display()
```

Non-Parameterized Constructor

A constructor without any arguments is called a non-parameterized constructor. This type of constructor is used to initialize each object with default values. This constructor doesn't accept the arguments during object creation. Instead, it initializes every object with the same set of values.

```
class Company:
    def __init__(self):
        self.name = "hello world institute"
        self.address = "Indore"
    def show(self):
        print('Name:', self.name, 'Address:', self.address)
cmp = Company()
# calling the instance method using the object
cmp.show()
```

Parameterized Constructor

A constructor with defined parameters or arguments is called a parameterized constructor. We can pass different values to each object at the time of creation using a parameterized constructor. We can pass different values to each object at the time of creation using a parameterized constructor.

The first parameter to constructor is self that is a reference to the being constructed, and the rest of the arguments are provided by the programmer. A parameterized constructor can have any number of arguments.



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```
class Employee:
    # parameterized constructor
    def __init__(self, name, age, salary):
        self.name = name
        self.age = age
        self.salary = salary
    def show(self):
        print(self.name, self.age, self.salary)
# creating object of the Employee class
emp1 = Employee('Emma', 23, 7500)
emp1.show()
emp2 = Employee('Kelly', 25, 8500)

emp2.show()
```

Constructor with Default Values

Python allows us to define a constructor with default values. The default value will be used if we do not pass arguments to the constructor at the time of object creation.

Example

```
class Student:
    # constructor with default values age and classroom
    def __init__(self, name, age=12, classroom=7):
        self.name = name
        self.age = age
        self.classroom = classroom
    def show(self):
        print(self.name, self.age, self.classroom)
Stu1 = Student('Emma')
Stu1.show()
Stu2 = Student('Kelly', 13)
Stu2.show()
```



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Constructor Overloading

Constructor overloading is a concept of having more than one constructor with a different parameters list in such a way so that each constructor can perform different tasks.

For example, we can create a three constructor which accepts a different set of parameters

Python does not support constructor overloading. If we define multiple constructors then, the interpreter will considers only the last constructor and throws an error if the sequence of the arguments doesn't match as per the last constructor.



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